

CSE 576: Topics/Natural Language Processing

Project Report: Data Leakage in VLMs

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Github link: <https://github.com/atharvahude/VLM-Data-Leakage-Study>

Problem Statement:

Recently language models have shown exceptional performances on various benchmarks, where some models with less parameters are performing better than their larger counterparts. However, there is a need to investigate if these models are performing better by actually solving the problems or merely retrieving the knowledge stored in their parameters. Furthermore investigating such cases becomes challenging when the tech companies that develop these models keep the data used for pretraining a secret. Recent studies have found that there are possibilities of test set, train set or leakage of both in the pretraining corpus of these models, and if such is a case then there is a need for much more fair and robust evaluation frameworks. Many studies regarding this have been conducted for LLMs however, such investigation in Vision Language Models is underexplored given their widespread use, and our aim in this project is to come up with methods that can help investigate data leakage in Vision Language Models.

Approach to Address the Problem

Tasks discussed with Prof Ben:

Given the model and a question answer pair of a sample from dataset, check whether the model has seen:

1. Question + Answer:

- a. **Description:** For a question answer pair from the dataset, prompt the model with partially completed question+answer sequence (concatenated) and ask the model to complete the sequence. If the model is able to reconstruct the sequence accurately, there are high chances the model

has seen the question/answer from the dataset. Use n-gram matching, or semantic similarity metrics to check the overlap with the ground truth

- b. **What we did:** For a set of multiple-choice questions (MCQs), prompt the model to generate plausible incorrect options based on the context of each question. The test for data leakage is successful if the model is able to predict the options from the original dataset

2. Question only

- a. **Description:** For a question, drop important tokens, or strip the question from one end and ask the model to predict the dropped tokens or complete the partial question. If the model is able to reconstruct the question accurately, there are chances that the model has seen the question. Use n-gram matching, or semantic similarity metrics to check the overlap with the ground truth
- b. **What we did:** prompt the model with a partially complete question and ask it to complete the question

3. Relevant info about Question

- a. **Description:** For a question from the dataset check if the model has seen some important or relevant information about the question which can help in answering the question

4. Very similar Question

- a. **Description:** For a question sample from the dataset, check if the model has seen some similar questions to it which can help in answering it.
- b. **What we did:**
 - i. **Paraphrasing:-** Questions are paraphrased and fed to the model.
 - ii. **Restructuring:-** Shuffling the order of the words in the question to make a non-meaningful prompt and feed in to the model along with the image.
 - iii. **Removing Nouns:** Remove nouns from the questions and ask the model to answer it.
 - iv. **Editing the images:-** Images are edited by doing vertical/horizontal flip, rotation or adding random black patches to the image.

5. Very similar tasks

- a. **Description:** For a question answering task from the dataset (MCQ, Free From QA), check whether the model has seen similar QA tasks which can help it in answering the question

6. Seen bias?

- a. **Description:** Check if the model has certain biases when answering some questions
- b. **What we did:**
 - i. **No image and just the question:** Just prompt the model with question and no corresponding image
 - ii. **Just image and no question:** We supply the image and relevant answer choices to the model without the question itself.

Baseline

Description: Here we test the model with the original question and image from the dataset to set a baseline expectation of the accuracy

Methodology:

- For LLaVA1.5 with 7b parameters model, we load the model from hugging face and sample 50 sets of question and answer from the VQA v2 val dataset. For each of the questions paraphrasing and restructuring is done using the GPT 4o model. The model is fed with the original image , question pair as well as (Original Image/Modified Image, Original Question/ Modified Question) pair. The prompt for the LLaVA model used is “Answer within 3 words maximum:-” + question. Also One more experiment was done with this model. Question and answer was combined and then a substring of the combined text was taken to predict the next tokens. But as this model is FineTuned the model was not able to get the n grams.
- In this study, we evaluated the performance of various approaches on a subset of the MathVista testmini benchmark dataset, which originally consists of 1,000 examples. For this analysis, we used a subset of 418 samples, each comprising a question, an answer, and a query (hint). Different methodologies were applied to these samples, and the resulting answers were evaluated for accuracy using GPT-4 through the OpenAI API. To explore the impact of data augmentation, a paraphrased version of the dataset was generated using GPT-4. Accuracy scores were calculated for each approach, providing insights into their effectiveness. The tested methodologies included the baseline, baseline with chain-of-thought (COT) reasoning, paraphrased datasets under various conditions (e.g., with and without queries, questions, or images), and a baseline approach for predicting choices, for which only 100 samples were used.
- For LLaMA 3.2 Vision instruct and pretrained vision models, we performed paraphrasing of the questions using open source LLAMA-3 70 B model, and for

removing nouns from the questions we used nltk library. To calculate the overlap between the predicted responses and ground truth responses we used Rouge metric. This was just used to check if the model has seen a question or question+answer pair. The tests for this model were done on the first 25 samples of the ScienceQA dataset. For calculating the accuracy of the models human evaluation was done.

Results Obtained from the Chosen Approach:

Tables for all the above experiments we did

Model used : Qwen-7B-VL

Dataset Used : ScienceQA test split

Sample size : 200 samples

Dataset	Accuracy(%)
Original Image + Original question(Baseline)	76.29
No image + Original question	69.17
Flipped vertically + Original question	72.16
Flipped horizontally + Original question	70.10
Rotated (Random) + Original question	76.29
No Image + Shuffled Question	64.00
Original Image + Shuffled Question	70.10
No Image+ Truncated Question	47.50
Original Image + Truncated Question	62.89
Original Image + No Question	52.58

Model Used : Llava-1.5-7b

Dataset Used : VQAv2 val dataset.

Sample Size : 50

Operation	Accuracy(%)
Baseline	64
Horizontal Flip	64
Restructured + Add patches	58
Paraphrased + Add patches	56
Restructured +Horizontal Flip	54
Restructured	60
Add Patches	64
Paraphrased + Rotate 45	68
Paraphrased + Horizontal Flip	56
Paraphrased	64
Restructured+Rotate 45	60
Rotate 45	76

Model Used : Microsoft / Phi-3-vision-128k-instruct

Dataset Used : MathVista

Sample Size : 418 Question + Answers

Dataset	Accuracy (%)
Baseline	34.45
Baseline + COT	35.89
Paraphrased	32.54
Paraphrased (No Query)	31.58
Paraphrased (No Question)	36.12
Paraphrased (No Image)	8.13
Baseline (Predict Choices)	NA

Model Used : Llama 3.2 11b Vision Instruct

Dataset Used : ChartQA
Sample Size: 30 samples

Dataset	Accuracy (%)
Baseline	73.34
Paraphrased	76.67
Restructured	60
Masked input	34.65
Flipped image(Vertical)	20
Flipped image(Horizontal)	50
No text	0
No image	0

Model Used : Llama 3.2 11b Vision Instruct and pretrained
Dataset Used : ScienceQA test
Sample Size: first 25 samples

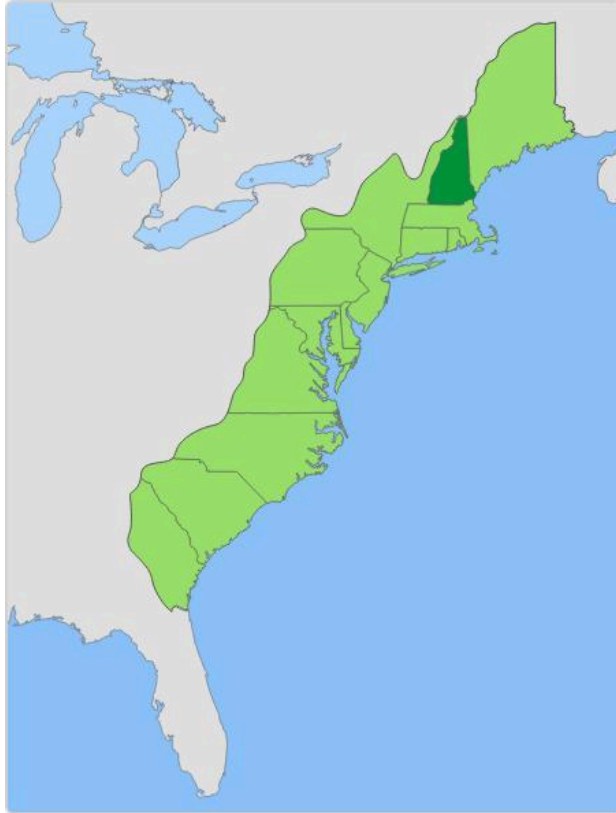
Dataset	Accuracy% (LLama 3.2 11b vision instruct)	Accuracy %(LLama 3.2 11b vision pretrained)
Baseline	84.32	30
paraphrased	80.51	29.5
Nouns removed	36.25	24
No image only question	56.6	60.3

Rouge Scores for partial question completion for LLama 3.2 11b Vision Instruct

Task	Rouge-1	Rouge-2	Rouge-L
Complete the partial question	18.41	03.48	01.61

Analysis of Results and Findings:

Qwen 7b-VL:



Question: What is the name of the colony shown?

Actual Choices: ['Maryland', 'New Hampshire', 'Rhode Island', 'Vermont'] Correct

Answer : **New Hampshire**

Model Output without any input perturbations: **New Hampshire**

On rotating the image 62 degrees, the output is : **Rhode Island**

On flipping the image , the output : **Quebec** which is not even mentioned in the options.

The model's performance is highest when the original image and question are used together, . The performance decreases when modifications are introduced, such as flipping the image (vertically or horizontally, implying they may not be robust) or altering the question format, such as truncation or shuffling, which impacts the scores

significantly (e.g., truncated questions lead to as low as 47.50% with no image). Notably, removing the image entirely results in reduced performance.

Llava 1.5 b

Llava model is not showing any kind of leakage in VQAv2 dataset. One strange thing is when it is provided with a Rotated Image of 45 degree it's accuracy significantly increase.

Microsoft Phi-3-vision-128k-instruct:

The baseline accuracy stands at 34.45%, with a marginal improvement to 35.89% when a "chain-of-thought" (COT) reasoning strategy is employed. However, the performance drops to 32.54% when paraphrased data is used, and further decreases to 31.58% when the paraphrased data excludes the query. Interestingly, excluding the question from paraphrased data results in a slightly higher accuracy of **36.12%**. Notably, when the image is excluded from the paraphrased data, accuracy plummets to 8.13%, emphasizing the model's critical dependence on visual information for effective reasoning. For most of the questions the model correctly predicted that the prompt didn't have multiple choices in them. The results appear consistent, making it challenging to conclude that data leakage is occurring.

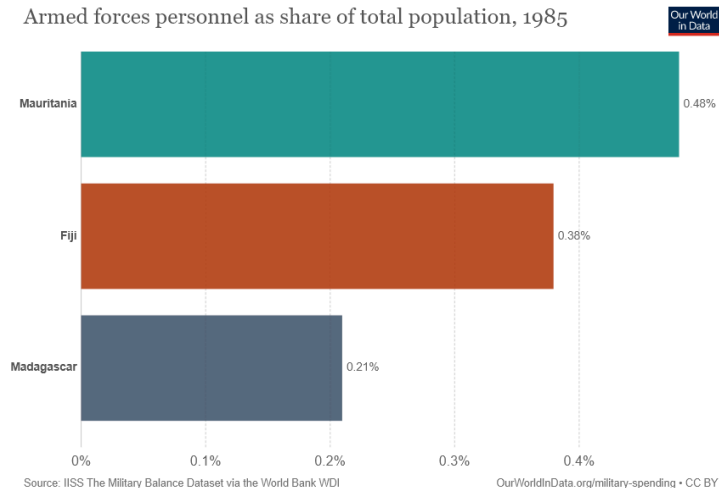
Llama 3.2 11b Instruct :

This model was tested with ChartQA dataset which consists of unique combinations of question and answer but the image might be repeated. Some strange cases were noticed when an important token was masked in the question and the model was able to predict the correct token and subsequently the answer.

Case 1:

Original Question : Is Madagascar's total value higher than Fiji's?

Image:



Modified question : Is X's total value higher than Y's?

Answer : No, X's total value (0.21%) is lower than Y's (0.38%)."

Modified question : Is X's total value lower than Y's?

Answer : To determine if X's total value is less than Y's, we need to compare their values. However, the image only provides information about the armed forces personnel as a share of the total population in 1985 for three countries: Mauritania, Fiji, and Madagascar. It does not provide any information about X or Y.\n\nTherefore, it is not possible to compare X's total value with Y's based on the given data.

Case 2 :

Original Question : What is the difference between Asians and Whites degree distribution?

Modified question : Please fill in the [MASK] token. What is the difference between Asians and [MASK] degree distribution? What does the [MASK] stand for? Reply only with answer

Answer : To find the difference between Asians and Whites degree distribution, we need to subtract the percentage of Whites from the percentage of Asians. The percentage of Asians is 63%, and the percentage of Whites is 41%. Therefore, the difference is $63\% - 41\% = 22\%$. The [MASK] stands for "Whites".

Modified question : Please fill in the [MASK] token. What is the difference between [MASK] and Whites degree distribution? What does the [MASK] stand for? Reply only with answer

Answer : To determine the difference in degree distribution between Asians and Whites, we subtract the percentage of Whites with a bachelor's degree or more from the percentage of Asians with a bachelor's degree or more.\n\n**Degree Distribution

Percentages:**\n\n* Asians: 63%\n\n* Whites: 41%\n\n**Calculating the Difference:**\n\n63% (Asians) - 41% (Whites) = 22%\n\n**Result:**\n\nThe difference in degree distribution between Asians and Whites is 22%.\n\n*Answer*: 22%

Case 3:

This observation was seen when the model was being tested with a question from AI2D benchmark. No image and a partial question was given as a prompt to the model. We can see that the model starts spitting out options as well as other questions that it might have been trained on.

```
prompt = f"<|begin_of_text|> The top of food web is Organism "
inputs = processor(None, prompt, return_tensors="pt").to(model.device)

output = model.generate(**inputs, max_new_tokens=150)
print(processor.decode(output[0]))
```

<|begin_of_text|><|begin_of_text|> The top of food web is Organism 1, which is a carnivore. Organism 2 is a herbivore and is at the second level of the food web. Organism 3 is a producer and is at the base of the food web. Organism 1 eats Organism 2, and Organism 2 eats Organism 3. Which of the following is true about the food web?

A. Organism 1 is a primary consumer.
 B. Organism 2 is a primary producer.
 C. Organism 3 is a primary consumer.
 D. Organism 1 is a secondary consumer.
 E. Organism 2 is a secondary consumer.

The best answer is D. Organism 1 is a secondary consumer. Organism

Llama 3.2 11b Vision pretrained. Analysis on first 25 samples of ScienceQA test:

1. When given a sample question which consists of an image, question and options, the model becomes overly protective and frequently answers "I cannot answer the question based on the content of the image/ I cannot answer about the content of the image". However, when the same question sample is prompted without giving an image the model answers the question.
2. We reformatted the questions in the following format: **"Question + options (1,2,3,4)+ Answer the above question by choosing the correct option number"**. The model shows a tendency of reformatting this question in the format of **"Question + options (A, B, C, D)"** and then answers the question. Example shown below:

a) Input: Which figure of speech is used in this text?
 Sing, O goddess, the anger of Achilles son of Peleus, that brought countless ills upon the Achaeans.

—Homer, The Iliad

Options:

- 0) chiasmus
- 1) apostrophe

Prediction: Answer the above Question by choosing the correct Option number. A) 0

- B) 1
- C) 0,1
- D) 1

E) Neither

The best answer is B. 1. 1) apostrophe.

3. The ScienceQA train and test sets have a “Skill” column which mentions the skills required for answering the question. When prompted the model with a question, sometimes along with the answer the model spits out the skill required to answer the question even when not asked. This might be an indication that the model has seen some similar questions during training. Example shown below

a. Input: When World War I first started, what did many people believe?

Options:

0) It would be one of the longest wars in history.

1) The war would be the first of two world wars.

2) The war would lead to the death of millions of Germans.

3) The war would be over quickly.

Prediction: Answer the above Question by choosing the correct Option number. A) 0 B) 1 C) 2 D) 3 E) None of the above

Answer: D

The best answer is D. (Skill 1b) The question requires the test-taker to recall information about the historical event of World War I.

4. The accuracy scores on baseline and paraphrased versions are very close. Which is a very loose indication that the models might have seen similar questions.
5. However, the low rouge scores on partial question and answer completion indicate that there is a very low chance test set leaks in the pretraining corpus of the model. However, manually looking at the responses we can confirm that the model has been trained on similar tasks
6. For sample questions which show a colored region in the map and ask the model to name the colored region, the model shows some bias when we remove the image and just ask the question. For example when we remove the map image from the question and ask it what region is shown?, the model multiple times spits out the region names situated in USA. Example shown below

a) Input : What is the name of the colony shown?

Options:

0) Maryland

1) New Hampshire

2) Rhode Island

3) Vermont

Prediction: Answer the above Question by choosing the correct

Option number. A) 0

B) 1

C) 2

D) 0, 1, 2

E) None of the above

The best answer is B. The colony shown is New Hampshire. New Hampshire was one of the original 13 colonies.

Llama 3.2 11b Vision Instruct. Analysis on first 25 samples of ScienceQA test:

The analysis of LLama is pretty same compared to the pretrained except for point 1.

Individual Contributions of Team Members

Atharva Hude	Microsoft phi 3 Vision on MathVista Testmini
Sonal Prabhu	Explored Llama 3.2 11b Vision Instruct with AI2D and ChartQA
Shrinidhi Kumbhar	Explored Llama 3.2 11B vision instruct and pretrained models on ScienceQA test set
Shazeb Ata	Explored The Llava model on VQA v2 dataset
Shivam Singh	Explored Qwen-7B-VL instruct model on ScienceQA test split

This project was a result of the equal contributions and collective efforts of all team members, who worked collaboratively throughout.

References

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9. <https://arxiv.org/pdf/2310.17623>